

● Original Contribution

QUANTIFICATION OF VAPORISED TARGETED NANODROPLETS USING HIGH-FRAME-RATE ULTRASOUND AND OPTICS

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Abstract—Molecular targeted nanodroplets that can extravasate beyond the vascular space have great potential to improve tumor detection and characterisation. High-frame-rate ultrasound, on the other hand, is an emerging tool for imaging at a frame rate one to two orders of magnitude higher than those of existing ultrasound systems. In this study, we used high-frame-rate ultrasound combined with optics to study the acoustic response and size distribution of folate receptor (FR)-targeted versus non-targeted (NT)-nanodroplets *in vitro* with MDA-MB-231 breast cancer cells immediately after ultrasound activation. A flow velocity mapping technique, Stokes' theory and optical microscopy were used to estimate the size of both floating and attached vaporised nanodroplets immediately after activation. The floating vaporised nanodroplets were on average more than seven times larger than vaporised nanodroplets attached to the cells. The results also indicated that the acoustic signal of vaporised FR-targeted-nanodroplets persisted after activation, with 70% of the acoustic signals still present 1 s after activation, compared with the vaporised NT-nanodroplets, for which only 40% of the acoustic signal remained. The optical microscopic images revealed on average six times more vaporised FR-targeted-nanodroplets generated with a wider range of diameters (from 4 to 68 μm) that were still attached to the cells, compared with vaporised NT-nanodroplets (from 1 to 7 μm) with non-specific binding after activation. The mean size of attached vaporised FR-targeted-nanodroplets was on average about threefold larger than that of attached vaporised NT-nanodroplets. Taking advantage of high-frame-rate contrast-enhanced ultrasound and optical microscopy, this study offers an improved understanding of the vaporisation of the targeted nanodroplets in terms of their size and acoustic response in comparison with NT-nanodroplets. Such understanding would help in the design of optimised methodology for imaging and therapeutic applications. (E-mail: mengxing.tang@imperial.ac.uk) © 2019 World Federation for Ultrasound in Medicine & Biology. All rights reserved.

Key Words: Targeted molecular imaging, Nanodroplets, Phase-change contrast agents, Microbubbles.

INTRODUCTION

Medical ultrasound imaging is a highly accessible and affordable tool that can provide high-resolution and real-time images without ionising radiation. The advent of contrast-enhanced ultrasound (CEUS) with microbubble contrast agents, together with the development of contrast-specific pulse sequences, has significantly increased imaging sensitivity from within the blood (Eckersley et al. 2005). The microbubbles have been used in

ultrasound-mediated therapies as well (Dijkmans et al. 2004). Recently, the field has seen new developments in high-frame-rate CEUS imaging (Leow et al. 2015; Tremblay-Darveau et al. 2014).

It has been reported that low-boiling-point perfluorocarbon nanodroplets can be formed simply by condensing their perfluorocarbon microbubbles with increasing pressure and decreasing temperature (Sheeran et al. 2011). These low-boiling-point nanodroplets can be rapidly activated to microbubbles by the application of external acoustic pressure, commonly referred to as *acoustic droplet vaporisation* (Kripfgans et al. 2000). These nanodroplets have three main advantages compared with

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microbubbles in the context of CEUS imaging applications. Firstly, nanodroplets have smaller diameters than microbubbles, which enable them potentially to go beyond the vascular region not accessible to microbubbles. Secondly, nanodroplets have longer half-lives than microbubbles under *in vivo* conditions. Thirdly, nanodroplets can be activated into microbubbles in any preset time and location (Sheeran and Dayton 2014). Such features of nanodroplets have enabled their application for super-resolution imaging (Zhang et al. 2018a, 2018b, 2018c), dual-modality imaging (Lin et al. 2017a, 2017b), high intensity focused ultrasound (HIFU), therapeutic histotripsy (Aydin et al. 2016; Moyer et al. 2015), sonoporation (Liu et al. 2016), chemotherapy (Zhu et al. 2017) and drug and gene delivery (Zhou et al. 2014). In biology and tissue engineering, nanodroplets can also be used for DNA fragmentation (Kasoji et al. 2015) and acoustically responsive scaffolds (Moncion et al. 2016), whereas microbubble contrast agents cannot.

In cancer imaging and therapy, nanodroplets could play an important role as they are approximately three to six times smaller in diameter than microbubbles before vaporisation theoretically (Mercado et al. 2016; Oleksandr et al. 2013) and approximately 12-fold smaller in diameter experimentally (Sheeran et al. 2015). These nanodroplets have the potential to extravasate from the vascular endothelium to the cancerous tissue and form target-specific bonds. Once bound, the targeted nanodroplets can be vaporised into microbubbles for imaging and therapeutic purposes. The maximum size of particles extravasated from the vasculature is approximately 10 nm in normal tissue, whereas the gap junction on the porous tumour endothelium may be as large as 200 nm (Ji et al. 2013). The enhanced permeability and retention (EPR) effect increases the passive accumulation of nanoparticles (50–200 nm) in tumour tissue. It has been found that nanodrug delivery to tumors by the EPR effect can offer only a twofold improvement in nanodrug delivery compared with normal tissue, which results in an insufficient drug concentration for treating most cancers (Nakamura et al. 2016). With respect to cancer, it has been found that the folate receptor (FR) is overexpressed on the surface of breast, ovarian, cervical and colorectal cancer cells, but not in normal human tissue (Parker et al. 2005). The treatment of ovarian cancer in phase II clinical trials used a FR-targeted chemotherapeutic agent combined with FR-targeted single-photon emission computed tomography imaging agents (Morris et al. 2014; Naumann et al. 2013). In addition, recent work (Hadinger et al. 2018; Marshalek et al. 2016) using FR-targeted nanodroplets indicates that targeted agents can also selectively promote intracellular delivery, which can be incorporated as a strategy for using

nanodroplets for even more precise cancer cell imaging and therapy (Matsunaga et al. 2012).

The low-boiling-point decafluorobutane nanodroplets (boiling point = -2°C) have been investigated for imaging purposes (Zhang et al. 2017, 2018a, 2018b, 2018c). Ultrasound molecular imaging using low-boiling-point nanodroplets was first reported in 2013 (Sheeran et al. 2013). As proof-of-principle, Sheeran and co-workers used arginine–glycine–aspartic acid (RGD) peptide-targeted nanodroplets to target the overexpressed $\alpha_v\beta_3$ integrin on human umbilical vein endothelial cells. However, this study was performed using a clinical Siemens imaging system, and thus, the imaging frame rate was limited to 30 Hz. Using confocal microscopy, Marshalek et al. (2016) reported on the capability of low-boiling-point FR-nanodroplets to target MDA-MB-231 breast cancer cells, followed by internalisation *via* folate receptor-mediated endocytosis. However, some important aspects of the targeted nanodroplet immediately after vaporisation, including the acoustic response and size distribution of the resultant microbubbles are not fully understood.

In this study, we investigated the acoustic response and size distribution of vaporised folate receptor-targeted nanodroplets with MDA-MB-231 breast cancer cells using high-frame-rate ultrasound and optics. Taking advantage of high-frame-rate CEUS and optical microscopy, this study offers an improved understanding of the vaporisation of the targeted nanodroplets in terms of their size and acoustic response in comparison with non-targeted (NT)-nanodroplets. Such an understanding would help in the design of optimised methodology for imaging and therapeutic applications.

METHODS

Preparation and characterisation of nanodroplets

Preparation of nanodroplet solutions was adapted from previously described methods (Marshalek et al. 2016). For the NT-nanodroplet lipid solution, the lipid colloid was generated by dissolving a 9:1 molar ratio of 1,2-dipalmitoyl-*sn*-glycero-3-phosphocholine (DPPC) and 1,2-distearoyl-*sn*-glycero-3-phosphoethanolamine-*N*-[amino(polyethylene glycol)-2000] (DSPE-PEG-2000) into a solution of phosphate-buffered saline (PBS), propylene glycol and glycerol (16:3:1, v/v) to achieve a total lipid concentration of 1 mg/mL (Avanti Polar Lipids, Alabaster, AL, USA). For the FR-targeted lipid solution (FR-nanodroplets), 1, 2-distearoyl-*sn*-glycero-3-phosphoethanolamine-*N*-[folate(polyethylene glycol)-2000] (DSPE-PEG-2000–folate) was added to the mixture of lipid powder to maintain a 9:0.8:0.2 molar ratio of DPPC, DSPE-PEG-2000 and DSPE-PEG-2000–folate, respectively. The mixture of lipid powder was also dissolved into a solution of PBS, propylene glycol and glycerol (16:3:1, v/v)

to achieve a total lipid concentration of 1 mg/mL (Avanti Polar Lipids). The DSPE–PEG-2000–folate lipid bioconjugate was prepared according to a previous described method (Chen *et al.* 2013). A 1-mL volume of lipid solution was added to a 2-mL glass vial.

The headspace of the vial was purged with decafluorobutane gas (Fluoromed, Round Rock, TX, USA) via an inlet needle, and the content equilibrated with a vent needle. Mechanical agitation was applied using a modified amalgamator (Bristol-Myers-Squibb, New York, NY, USA) to produce the stable microbubbles.

For condensing microbubbles into nanodroplets, the headspace of the vial was pressurised according to previous methods (Zhang *et al.* 2018a, 2018b, 2018c). Briefly, the vials of microbubbles were immersed in an ice-salt bath (-5°C to -10°C), followed by pressurisation with ambient air into the vial septum.

A NanoSight NS300 (Malvern Instruments, Inc., Malvern, UK) was used to measure the size and concentration of both FR- and NT-nanodroplet solutions via nanoparticle tracking analysis (NTA), the nanoparticle measurement range being 10 to 1000 nm. Both nanodroplet solutions were diluted 1000-fold in de-ionised water. Three samples were prepared for each nanodroplet solution, and three measurements were performed for each sample. For both FR- and NT-nanodroplet solutions, nine measurements in total were performed to obtain the mean values of size and concentration of each nanodroplet solution. Five vials of nanodroplet solution were consolidated to form the final nanodroplet solution for the experiments.

Cell culture

The immortalised human breast cancer cell line MDA-MB-231 was obtained from an immortalised cell line maintained at the Hammersmith Hospital campus at Imperial College London. Cells were grown in RPMI-1640 medium supplemented with 10% fetal bovine

serum, 2 mM L-glutamine, 100 U/mL penicillin and 100 $\mu\text{g/mL}$ streptomycin (Sigma-Aldrich, St. Louis, MO, USA). Cells were cultured in a T75 culture flask at 37°C in an incubator with 5% CO_2 . The mean diameter of the cells ranged from about 20 to 30 μm as noted in prior literature (Kim *et al.* 2007).

Experimental setup

MDA-MB-231 cells were detached with 0.25% trypsin–ethylenediaminetetraacetic acid (Sigma-Aldrich). Detached cells were neutralised using full serum medium and centrifuged at 1300 rpm for 5 min. Cell pellets were resuspended in 1 mL of medium before seeding 2.2 million cells with 10 mL of medium in an optically and acoustically transparent Opticell chamber (Nalge Nunc International, Rochester, NY, USA). The inner dimensions of the Opticell chamber are $7.6 \times 6.6 \times 0.2$ cm, respectively. The thickness of the MDA-MB-231 breast cancer cell layer is normally about 5 μm according to previous literature (Winnard *et al.* 2008). After 24 h of incubation, 0.2 mL of nanodroplet solution ($\sim 1.1 \times 10^9$ nanodroplets) was added to the medium, and the diluted solution was mixed well before being injected into each Opticell chamber. The chamber was incubated at 37°C for 10 min. After incubation, medium was aspirated and cells were washed with additional medium three times to remove any residual unbound nanodroplets from the Opticell chamber. The Opticell chamber was then immersed and equilibrated in a 37°C temperature-controlled water tank. The transducer was held 15 mm above the top surface of the Opticell chamber, as illustrated in Figure 1. Acoustic absorbers were placed at the bottom of the water tank to reduce ultrasound reflection. Immediately after the acquisition of ultrasound data, the Opticell chambers were transferred to acquire microscopic images, obtained using a Nikon DXM-1200C (Nikon Instruments Inc.) equipped with a $40\times$ objective lens and a numerical aperture of 0.65. The resolution of the microscope is $0.1613 \mu\text{m/pixel}$. The microscopic imaging plane was focused directly on the breast

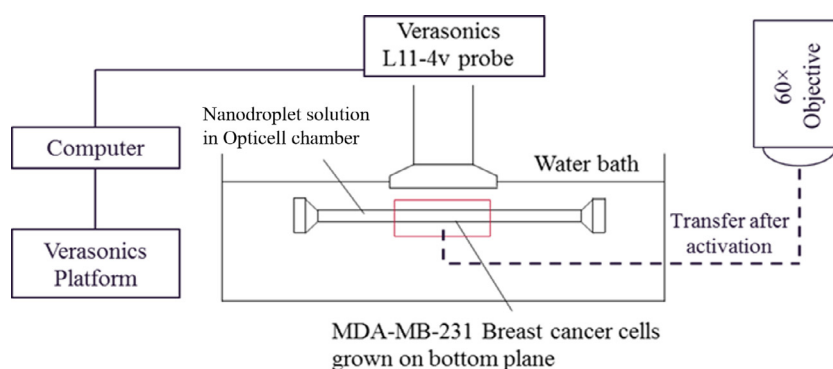


Fig. 1. Schematic of the experimental setup (not to scale). The red box represents the region of interest of ultrasound images at the upper and lower planes (MDA-MB-231 breast cancer cells) of the Opticell chamber.

Table 1. Imaging—activation—imaging sequence and the corresponding ultrasound parameters used in the experiments

State	Pulse type	Transmit frequency (MHz)	Pulse length (half-cycle)	PNP (MPa)	Mechanical index
Imaging (before)	Plane wave	4	1	0.20	0.10
Activation	Focus wave	8	10	5.37	1.90
Imaging (after)	Plane wave	4	1	0.20	0.10

PNP = spatial and temporal peak-negative-pressure.

cancer cells fixed to the bottom plane of the Opticell chamber. As a reference point for microscopic imaging, the Opticell chamber under the transducer where ultrasonic activation was conducted was marked. For each Opticell chamber, five independent optical measurements in different regions were acquired before and after ultrasound exposure to observe the newly vaporised nanodroplets attached to the cell surface. Two Opticell chambers were used for FR-nanodroplets and another two for NT-nanodroplets.

Imaging acquisition and analysis

A customised “imaging—activation—imaging” acquisition sequence, as outlined in Table 1, was implemented on an ultrasound research platform (Verasonics Vantage 128, Kirkland, WA, USA) with a L11-4 38-mm linear array probe (ATL, USA) (Lin et al. 2016). The spatial and temporal peak-negative pressures (PNPs) were measured by hydrophone calibration as described by Lin et al. (2017a, 2017b). The acoustic vaporisation of nanodroplets was achieved by electronically sweeping the transmitted focus waves (8-MHz, 5-cycle sinusoidal waves with PNP of 5.37 MPa, f -number = 0.4, axial focus = 15 mm) over the depth of interest. Only one activation focus pulse was transmitted to activate the nanodroplets. For imaging, a coherent compounding plane wave pulse-inversion technique was employed to form an image (Montaldo et al. 2009). Plane waves were transmitted at nine different angles (angle range: -10° to 10°) at 4 MHz with a PNP of 0.20 MPa. The ultrasound imaging resolution cell size was $0.56 \times 0.64 \times 1.5$ mm, respectively, in the axial, lateral and elevational directions. Similar to conventional pulse-inversion technique, a “positive” pulse and a “negative” pulse were acquired at each transmission angle to suppress tissue signal and enhance microbubble signal (Shen et al. 2005). Fifty pre-activation and 500 post-activation images were collected at a rate of 500 frames/s. A differential imaging technique was applied to remove the background signal (Brown et al. 2017). Frames after activation were subtracted by the average of frames before activation to remove background signals. The region of interest was defined by the bottom plane (indicated by the blue box between 16 and 17 mm in Fig. 2) of the Opticell chamber, where MDA-MB-231 breast cancer cells were attached. Customised MATLAB (The MathWorks, Natick, MA, USA) scripts were written to enable all the analyses and

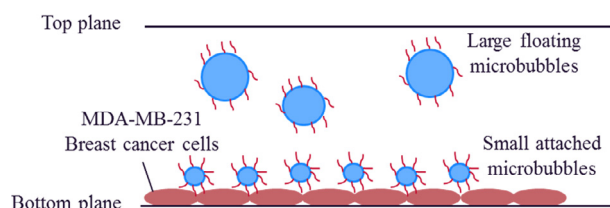


Fig. 2. Schematic of the cross section of an Opticell chamber after acoustic activation of folate receptor-targeted nanodroplets. A portion of newly vaporised folate receptor-targeted nanodroplets were still attached to breast cancer cells, whereas another portion of large vaporised folate receptor-targeted nanodroplets were floating.

calculations. All images are shown in gray scale, where each pixel in the image is normalised by the maximum pixel value.

Flow velocity mapping and size estimation

After ultrasonic exposure of FR-nanodroplets, a portion of resultant microbubbles were still attached to the cells, whereas another portion of large bubbles floated up because of buoyancy, as illustrated in Figure 2.

The rising velocity of the vaporised nanodroplet (microbubble) in the stagnant liquid is governed mainly by the buoyancy force and the drag force experienced by the vaporised nanodroplets. In the force balance approach, the rising velocity is calculated from the balance of buoyancy and drag force. The rising velocity of vaporised nanodroplets in a viscous liquid at low Reynolds number can be expressed by the equation

$$V = \frac{(\rho_l - \rho_g)gD_b^2}{18\mu_l} \quad (1)$$

Calculations were adapted from Stokes' equation (Parmar and Majumder 2015), where V is the rising velocity, D_b is the diameter of the microbubble, ρ_l is the density of liquid at 37°C (approximately 1 g/cm^3), ρ_g is the density of gas at 37°C (11.2 kg/m^3), g is the gravity acceleration (9.8 m/s^2) and μ_l is the viscosity of liquid at 37°C ($8.90 \times 10^{-3} \text{ dyn s/cm}^2$).

The size distribution of the attached vaporised nanodroplets can be directly estimated from microscopic images. Individual vaporised nanodroplets were segmented out of the images to estimate the diameter distribution of both

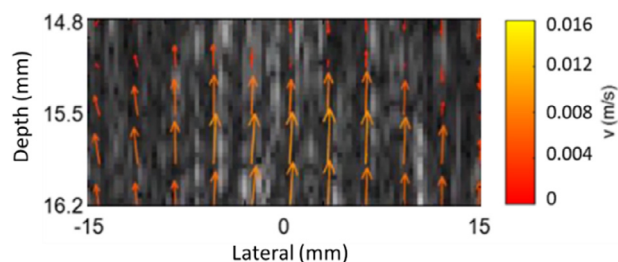


Fig. 3. Quantitative flow vector visualization of vaporized FR-nanodroplets generated immediately after activation inside the Opticell chamber. Each measurement is averaged over 10 ms to reduce the noise. The data used is after the processing of singular value decomposition filtering to reduce the noise. The length and direction of the vector arrow represent the magnitude and direction of the velocity within the region of the arrow.

vaporised FR- and NT-nanodroplets that remain attached to the cells. A customised MATLAB script was written to perform the segmentation to separate the microbubbles out and measure their diameters. The size distribution of floating vaporised nanodroplets can be estimated by tracking average rising velocities of the newly vaporised nanodroplets immediately after acoustic activation using the flow velocity mapping technique described previously (Leow *et al.* 2015) combined with Stokes' equation (Parmar and Majumder 2015). Briefly, the cross-correlation analysis was performed to compute a local velocity between the corresponding interrogation windows of two neighbouring ultrasound frames. A velocity map was displayed with color-coded velocity vectors. The full flow field within the imaging plane of the Opticell chamber was estimated, and thus flow velocity within any point in the imaging plane could be determined. The number of vectors and their locations in the image/video were chosen for visualisation purposes. By estimating the average flow velocity in the resolution cell in all individual trials, the average size of the bubble population could be obtained *via* Stokes' equation.

In Figure 3 is representative flow velocity mapping of vaporised nanodroplets immediately after activation (Supplementary Video S1, online only). The movement of the vaporised nanodroplets could be visualised, and the corresponding average flow velocity could be obtained. The average flow velocity obtained from flow velocity mapping was used in combination with eqn (1) for the size estimation.

Statistical analysis

All data are expressed as the mean $\pm$ standard deviation. Statistical significance was analyzed using the two-sample Student *t*-test (with the assumption of *t*-distribution) and two-sample *z*-test (with the assumption of normal distribution) according to the

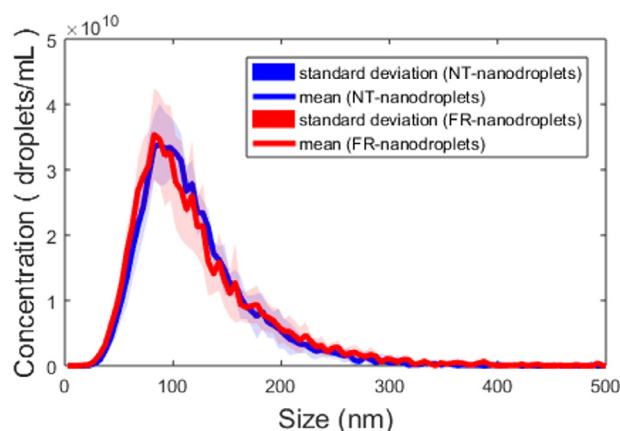


Fig. 4. Concentration and size characterisation of non-targeted and folate receptor-targeted nanodroplets measured using nanoparticle tracking analysis ($N=9$). The curve and its corresponding shaded error bar represent the mean and standard deviation, respectively. Non-targeted nanodroplets: 118.7 ± 6.7 nm, folate receptor-targeted nanodroplets: 116.8 ± 5.9 nm.

sample size. Student's *t*-test was used to compare two sets of data with sample size <30 , and the two-sample *z*-test was used to compare two sets of data with sample size >30 . A *p* value < 0.05 was considered to indicate statistical significance; a *p* value < 0.01 , strong significance; and a *p* value < 0.001 , high significance.

RESULTS

Characterisation of nanodroplets

The size distributions of NT-nanodroplets and FR-nanodroplets measured using NTA are illustrated in Figure 4. No statistically significant differences were found between the mean diameters of NT-nanodroplets and FR-nanodroplets ($N=9$). The mean diameters of NT-nanodroplets and FR-nanodroplets were 118.7 ± 6.7 and 116.8 ± 5.9 nm, respectively, and were within the size range for promoting the EPR effect (≤ 400 nm) reported by the previous study (Ernsting *et al.* 2013). The concentrations of NT- and FR-nanodroplets were $(3.51 \pm 0.26) \times 10^{10}$ and $(3.45 \pm 0.31) \times 10^{10}$ droplets/mL, respectively.

Acoustic response of vaporised nanodroplets

In Figure 5 are temporal snapshots of FR- and NT-nanodroplets in the Opticell chamber 0.002 s before and 0.002 and 1 s after activation (Supplementary Video S2, online only). The bottom plane where the cells were located was between 16 and 17 mm. More persistent and visible ultrasound contrast signal could be seen for vaporised FR-nanodroplets than for vaporised NT-nanodroplets 1 s after activation.

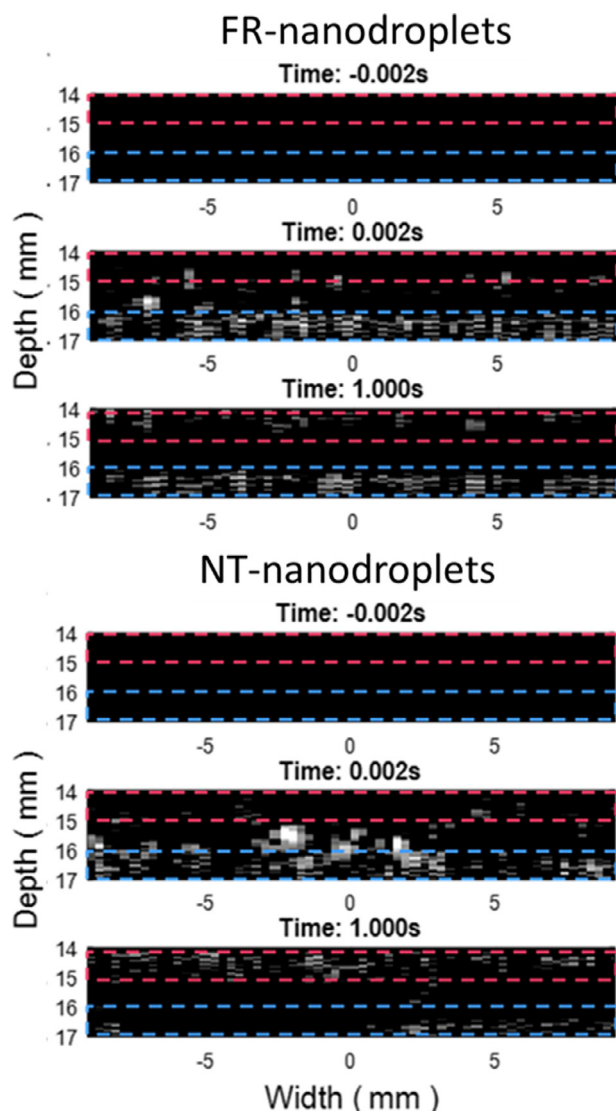


Fig. 5. Representative temporal snapshots show FR- and NT-nanodroplets in Opticell chamber 0.002s before, and 0.002s and 1s after activation respectively using the pulse inversion imaging at MI=0.1 and focus-pulse activating at MI=1.9. The region of interest (ROI) shows the top plane (indicated by the red box between 14 to 15 mm) and bottom plane (indicated by the blue box between 16 to 17 mm) of the Opticell chamber where MDA-MB-231 breast cancer cells located.

Figure 6 illustrates the quantitative analysis of acoustic response (*i.e.*, gray-scale value) of vaporised FR- and NT-nanodroplets. First, the normalised contrast amplitude of the FR-nanodroplets is significantly higher than that of the NT-nanodroplets. Second, the contrast signal of FR-nanodroplets was relatively persistent after activation, with 70% of the acoustic signals remaining 1 s after activation, compared with the NT-nanodroplets, for which only 40% of the contrast signals remained. The contrast signal from NT-nanodroplets was fast compared with that from FR-nanodroplets.

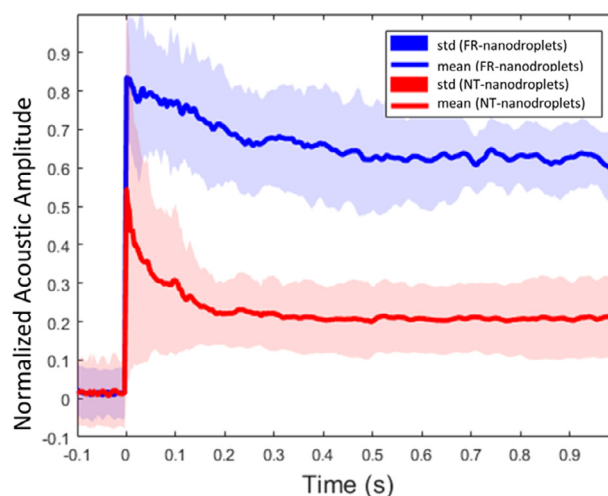


Fig. 6. Quantification of normalized acoustic amplitude (*i.e.* the grayscale value of pixels in ROI) of activation of FR- and NT-nanodroplets within bottom ROIs in the Opticell chambers using the pulse inversion imaging at MI=0.1 and focus-pulse activating at MI=1.9. The blue curve represents for FR-nanodroplet and the red curve represents for NT-nanodroplet. The shaded error bar represents the corresponding standard deviation.

Size comparison between floating and attached vaporised nanodroplets

In Figure 7 are the acquired microscopic images focused at the bottom of the Opticell chamber before and after ultrasound exposure. Figure 8 illustrates the quantification of size distribution of both vaporised FR- and NT-nanodroplets attached to the cells. The number of vaporised FR-nanodroplets observed on the bottom plane was 6.3 times larger than that of vaporised NT-nanodroplets. The vaporised FR-nanodroplets had significantly larger diameters (mean = 9.8 μm) compared with the vaporised NT-nanodroplets (mean = 3.6 μm). Moreover, the vaporised attached FR-nanodroplets had a diameter distribution between 4 and 68 μm , whereas the vaporised attached NT-nanodroplets had a much narrower range (from 1 to 7 μm). Although some small vaporised NT-nanodroplets could be seen, they were not adherent to the cells according to Figure 7.

The size distributions of attached vaporised nanodroplets were obtained using microscopic imaging and microbubble segmentation (Sennoga et al. 2010), whereas the size distributions of floating vaporised nanodroplets were obtained using ultrasound imaging, flow velocity mapping and Stokes' theory. These two sets of data are illustrated in a boxplot in Figure 9 for comparison. The mean diameters of vaporised FR- and NT-nanodroplets were 70 and 75 μm , respectively. The maximum diameter estimated was approximately 150 μm . The average size of floating vaporised nanodroplets was significantly larger than that of cell surface-bound microbubbles generated in both cases.

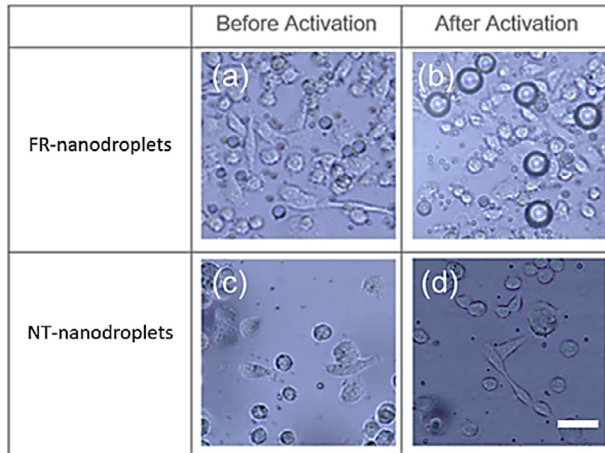


Fig. 7. Representative microscopic imaging of FR-nanodroplets and NT-nanodroplets before and after activation in Opticell chamber. (a) FR-nanodroplets before activation (b) FR-nanodroplets after activation (c) NT-nanodroplets before activation (d) NT-nanodroplets after activation. The scale bar represent 30 μm .

However, for the attached vaporised FR-nanodroplets, the maximum diameter of all microbubbles measured on the cell membrane was 68 μm , as observed in Figure 7.

DISCUSSION

Targeted nanodroplets for molecular imaging have potential in a wide range of clinical and pre-clinical applications in cancer and cardiovascular diseases (Marshalek *et al.* 2016; Porter *et al.* 2016). Using high-frame-rate CEUS and optical microscopy, this study offers an improved understanding of the vaporisation of the targeted nanodroplets in terms of their size and acoustic response in comparison with NT nanodroplets. Such an understanding will help in the design of optimised methodology for imaging and therapeutic applications.

For imaging applications, the bio-effects induced after the activation of nanodroplets are required to be minimal for safety considerations. Any vaporised droplets with a diameter larger than, for example, 10 μm , could cause occlusion in the microcirculation and increase the risk of cavitation-induced bio-effects and should be avoided (Sheeran *et al.* 2016a, 2016b). For therapeutic applications such as HIFU and histotripsy, the larger size of activated nanodroplets can facilitate cavitation to achieve better therapeutic performance (Moyer *et al.* 2015; Vlaisavljevich *et al.* 2017). Therefore, the size distribution of the nanodroplets is important for each specific application.

Comparison with the existing literature

The binding efficiency of the FR-nanodroplets in this study was about 2.4 activated nanodroplets per cell

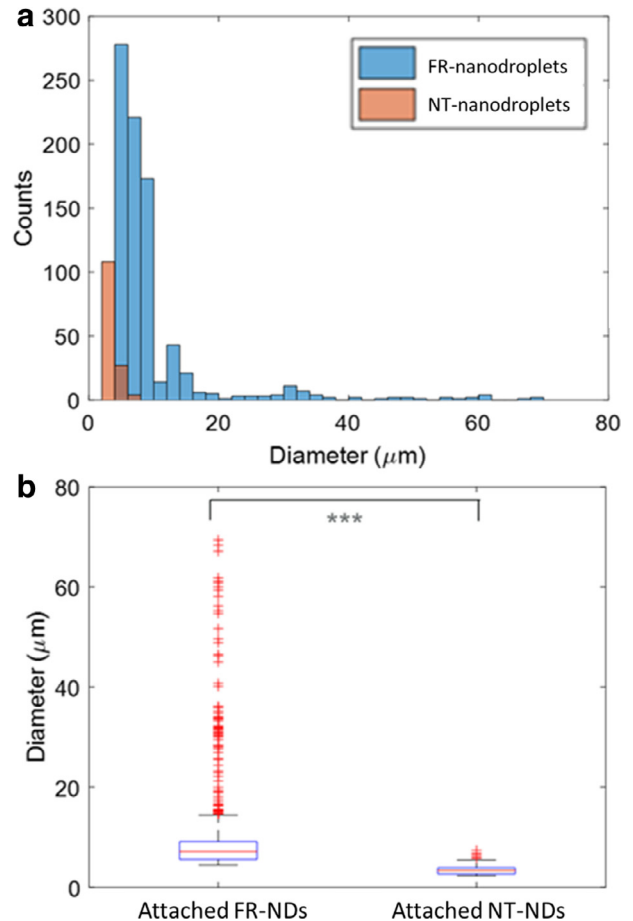


Fig. 8. The histogram (a) and box plot (b) shows the diameter distribution of both vaporized FR- (N=819) and NT- (N=139) nanodroplets (NDs) immediately after acoustic activation of nanodroplets. The newly vaporised FR-nanodroplets have a wider range of diameters than vaporized NT-nanodroplets. The mean diameter of vaporized FR-nanodroplets is significantly larger than that of vaporized NT-nanodroplets.

after activation according to the experimental results. For the size of targeted vaporised nanodroplets, Marshalek and co-workers (2016) used the same nanodroplets to target the MDA-MB-231 breast cancer cells; however, that work only focused on the size of the vaporised nanodroplets internalised within the cells, rather than those attached to the cell surface. Sheeran *et al.* (2013) used RGD-targeted nanodroplets to target human umbilical vein epithelial cells and found that the size of vaporised targeted nanodroplets ranged from 1 to 3 μm . In our study, a wide range of diameters (from 4 to 68 μm) of vaporised targeted nanodroplets could be observed in microscopic images. This could be due to the concentration used or the ultrasound pulse sequence employed, as well as the different targeted ligands used. Future studies on such effects need to be conducted. For the size of floating vaporised nanodroplets dislodged

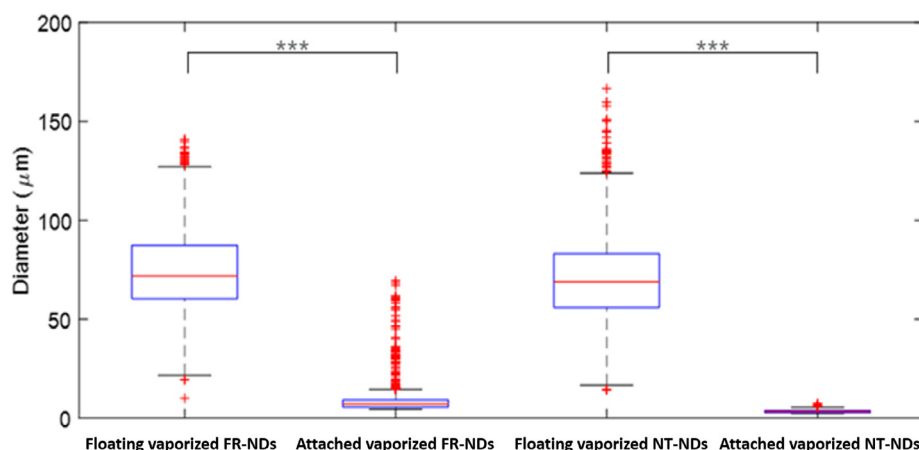


Fig. 9. The boxplot illustrates the size distribution of floating and attached FR-targeted and NT vaporised nanodroplets immediately after acoustic activation. The sizes of attached, vaporised nanodroplets were obtained from microscopic imaging. Alternatively, the sizes of floating vaporised nanodroplets were obtained from flow velocity mapping of ultrasound imaging. As can be seen from the figure, the size of floating vaporised nanodroplets were significantly larger than that of attached vaporised nanodroplets for both vaporised FR-NDs and NT-NDs. ***Highly significant difference, $p < 0.001$. FR-NDs = folate receptor-targeted nanodroplets; NT-NDs = non-targeted nanodroplets.

from cells, there is no previous literature on this topic to the best of our knowledge.

Different targeting ligands may have different affinities to the receptors on cells (Toporkiewicz et al. 2015). If one targeting ligand has a very high affinity to the receptor, there will be more droplets attached to cells before activation. The high concentration of bubbles generated by ultrasound-mediated activation could likely lead to diffusion/coalescence on the membrane surface. However, the assumption was made that diffusion/coalescence could as well occur on the cell surface. Therefore, the size may also change because of the different targeting ligands.

Acoustic response immediately after activation

A sharp decrease in the acoustic signal during the first 100 ms after activation is illustrated in Figure 6, likely because the large, newly vaporised nanodroplets float to the top plane of the Opticell chamber when experiencing buoyant force. As vaporised nanodroplets moved away from the region of interest as illustrated in Figure 5, there was a significant decrease in the acoustic signal. The populations of newly vaporised nanodroplets underwent some dynamic processes such as gas diffusion in or out of the microbubbles and the coalescence of microbubbles within the first 100 ms (Lajoinie et al. 2014; Raymond et al. 2016). Moreover, the formation of the vortex resulting from the high-mechanical-index activation pulse could also push the generated microbubbles out of the imaging plane, as there was no confinement in the elevational direction in the Opticell chamber (Raymond et al. 2016). The acoustic signal of newly vaporised NT-nanodroplets does not disappear at the end

of 1 s post-activation, mainly because the NT-nanodroplets are subjected to non-specific binding to cells. It is feasible that the lipid component could possibly infuse into the cell membrane bilayers to contribute to the non-specific binding (Escribá et al. 2008). Moreover, the acoustic signal decay may also be related to the cancer cell studied, the target ligand chosen (Marshalek et al. 2016), the activation and imaging pulses (Sheeran et al. 2013) and the confinement of experimental environment (Lin et al. 2017a, 2017b), as the Opticell chamber is a confined environment.

Size distribution of attached vaporised nanodroplets

After activation, vaporised FR-nanodroplets were expected to remain on the bottom plane because of FR binding to the cell membrane. Although the successful binding and vaporisation could be realised using nanodroplets, the maximum diameter of vaporised nanodroplets attached to the cells was as high as 68 μm , which was much larger than the mean diameter of the microbubble population before condensation (which has a peak size distribution centered approximately around 0.5–6 μm). The large FR microbubbles (68 μm) were bound to the targeted cells despite the relatively large buoyant force. There are a few possible explanations for the large newly generated microbubbles. First, from a chemistry perspective, the monolayer of the lipid shell will be zipped to form bilayer strands from the microbubble condensation to nanodroplets, followed by the unzipping of the bilayer strands after activation to re-form the monolayer (Mountford et al. 2015; Sheeran et al. 2016a, 2016b). The unstable vaporised nanodroplets may lack some lipids immediately after acoustic activation and subsequently may tend to coalesce

to share more lipid shell to promote the formation of larger microbubbles. In addition, a larger activation energy barrier is required to achieve the homogeneous nucleation of those nanodroplets of smaller diameters (<100 nm) (Mountford *et al.* 2015); thus, the nanodroplet size outliers (diameters >1000 nm) are more easily activated to form larger vaporised nanodroplets. Second, from an imaging perspective, the high-frame-rate ultrasound imaging pulse from the transducer may “push” the microbubbles to coalesce into larger microbubbles (Postema *et al.* 2004). The secondary radiation force may also affect microbubble coalescence, as described in the literature (Antonios and James 2016; Himanshu *et al.* 2017). Finally, from an activation perspective, although the activation frequency (8 MHz) used in this study was higher than the activation frequency used in some previous studies (Sheeran *et al.* 2015, 2016a, 2016b), it may still not be high enough to completely vaporise all the nanodroplets, especially those that are small and require higher frequencies to generate a “superharmonic focusing” effect to initiate the nanodroplet nucleation (Shpak *et al.* 2014).

Moreover, the mean size of attached vaporised FR-nanodroplets was about 3 times larger than that of attached vaporised NT-nanodroplets. This may be because more FR-nanodroplets attached to the MDA-MB-231 cells than NT-nanodroplets before ultrasound activation. Thus, after activation, there were more vaporised FR-nanodroplets undergoing diffusion/coalescence to form larger microbubbles.

Although aggregation of nanodroplets after their injection into the Opticell chamber cannot be absolutely ruled out, the nanodroplet membranes do contain polyethylene glycol-2000, which has a tendency to retard this phenomenon. Moreover, the aggregation of both kinds of nanodroplets was not observed under the microscope before activation, as the resolution of the microscope is $0.1613\text{ }\mu\text{m/pixel}$.

Spontaneous vaporisation of nanodroplets

Spontaneous vaporisation of FR-nanodroplets could be seen before ultrasonic activation. This is due to absorption of the thermal energy (temperature = 37°C) provided by the chamber environment. In this study, a high acoustic pressure (mechanical index = 1.9) activation pulse (with 10 half-cycles) was expected to destroy the existing spontaneously vaporised nanodroplets (William *et al.* 2006) and also to generate new ultrasound-activated nanodroplets. Hence, most of the vaporised nanodroplets observed after ultrasound activation were expected to be the vaporised nanodroplets activated by the ultrasound pulse.

There were possible interactions between the spontaneously vaporised nanodroplets and nanodroplets under ultrasound. Thus, the presence of these spontaneously vaporised nanodroplets could influence the

activation process of the nanodroplets in their proximity by potentially assisting in the activation of nanodroplets.

It is possible that a small amount of spontaneously vaporised droplets persist after activation (attached/dislodged). However, because this would occur for both FR- and NT- nanodroplet experiments, a significant difference in the size of the droplets is not expected.

A “no activation pulse” control experiment was performed and compared with an experiment with activation pulses, and the results are illustrated in Figure 10. As can be seen from the figure, the low-pressure imaging pulses were found not to be able to destroy any of the spontaneously vaporised droplets. For the experiment with imaging and activation pulses, as indicated by the *red arrows* in the figure, it can be clearly seen that the spontaneously vaporised droplets were destroyed by the activation pulses.

Flow velocity mapping

With respect to flow velocity mapping, it was difficult to associate the velocity vectors with individual vaporised nanodroplets because of the large size of the resolution cell, which might contain a number of vaporised nanodroplets. The signal amplitude within such a resolution cell is weighted by the individual bubble signal strengths, which may be highly size dependent. Furthermore, it could be potentially affected by a number of elements, such as complex function of scattering, resonance and non-linear detection scheme effects. However, we believe the size estimation approach used in this study is still able to determine the average size range of the vaporised targeted droplets and help us to understand their behavior after activation.

The imaging frequency used in this study was 4 MHz, which may not visualise well the small bubbles generated. The generated bubbles of approximately $2\text{ }\mu\text{m}$ may resonate at this imaging frequency according to a previous study (Tang and Eckersley 2006).

Assumption of Stokes' equation

The Stokes' equation was used as a mathematical model to convert the rise velocity to the diameter of rising microbubbles (Parmar and Majumder 2015). There are three assumptions for applying this equation. First, it is assumed that the microbubbles are perfectly spherical. Second, it is assumed that there is no internal gas circulation within the microbubbles and no slip exists at the boundary. Third, it is assumed that the liquid inside the chamber is stagnant. In this study, the activation pulse may induce a flow inside the chamber. However, as can be seen in Figure 3, the flow was still dominated by the rising microbubbles. It may be possible that some non-gaseous particles may have detached from the bottom of the Opticell chamber with the floating microbubbles;

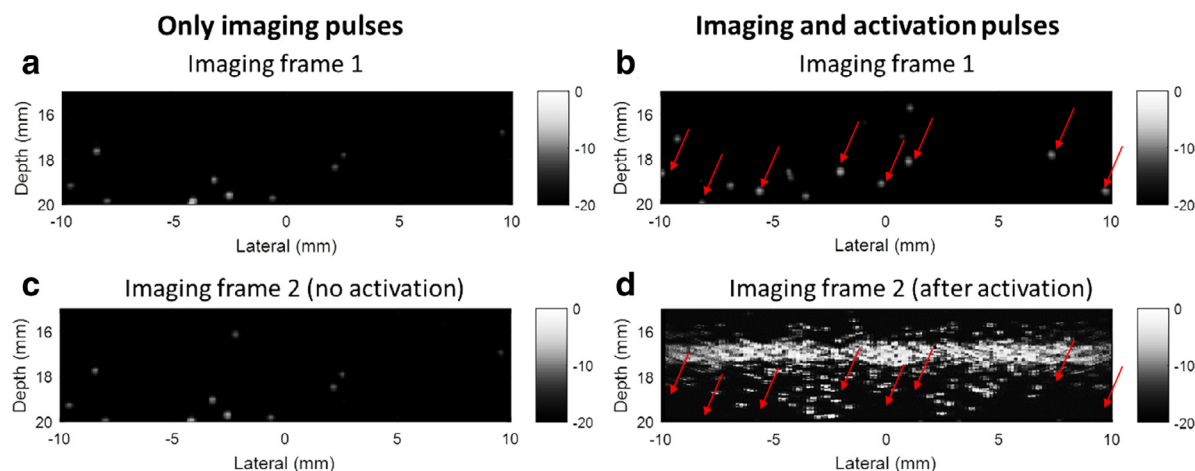


Fig. 10. A control experiment with only imaging pulses (left) and an experiment with both imaging and activation pulses (right) were performed at 37°C in a droplet solution, illustrating that the activation pulse was able to remove the spontaneously vaporised droplets. (a, c) Two consecutive imaging frames without activation pulses. (b, d) Two consecutive imaging frames (before and after activation, respectively), with activation pulses inbetween. The experimental conditions were kept the same as detailed in Table 1. Note that (a) and (b) are the very first frames of the individual experiment with spontaneously vaporised droplets, and the horizontal line in (d) corresponds to the ultrasound focal depth of the vaporisation pulses.

however, no non-gaseous attachment to the bubbles can be observed.

Size of nanodroplets before activation

Unfiltered (without size manipulation) FR- and NT-nanodroplets were used in this study. It is noted that the presence of nanodroplet size outliers may potentially affect the ability of acoustic vaporisation of nanodroplets to yield larger vaporised nanodroplets (Sheeran et al. 2016a, 2016b). In the *in vivo* setting, the large vaporised nanodroplets should be filtered out by the lungs (Butler and Hills 1979). Therefore, the size population of nanodroplets will need to be manipulated in future studies. Although the mean size of vaporised FR-nanodroplets generated immediately after activation is 9 μm , vaporised droplet outliers as large as 68 μm were detected. The size population of newly generated microbubbles will also be affected by the gas concentrations in the ambient fluid and confined environment (Lin et al. 2017).

MDA-MB-231 breast cancer cells

Having gone through a number of passages, MDA-MB-231 cells are immortalized. This large number of passages could result in cell receptor heterogeneity and, thus, different numbers of receptors. Although this feature has not been studied in detail, Marshalek et al. (2016) alluded to efficiency of binding to MDA-MB-231 cells using folate-targeted nanodroplets. This has also been noted for other cell lines (*i.e.*, pancreatic cancer [Panc]-1). Whether this is due to cell contact/confluence or a large number of passages generating heterogeneity is not explicitly known.

Ultrasound pulses on detachment

It has been reported that ultrasound pulses with even medium amplitude are able to detach targeted microbubbles (Loughran et al. 2012). Unlike imaging of vascular endothelial targets, where shear stress from blood flow can detach the microbubbles from their bound sites, reducing ultrasound signal, the flow is very limited outside the vasculature, and any detached bubbles could potentially stay in position and continue to contribute to ultrasound signal. The activation pulse could also cause diffusion of gas out of the microbubbles, resulting in loss of echogenicity. Future work is required to study the impact of potential detachment of vaporised nanodroplets on the ultrasound molecular images.

CONCLUSIONS

We have investigated binding of targeted nanodroplets to MDA-MB-231 breast cancer cells, as well as the acoustic response and size distribution of newly generated microbubbles immediately after activation. It was found that the size of floating microbubbles is on average more than seven times larger than the size of microbubbles targeted to the cells. The results reveal that the acoustic signal of generated FR-microbubbles persists after activation, with 70% of acoustic signal remaining 1 s after activation, compared with the NT-microbubbles, for which only 40% of the acoustic signal remains. The microscopic imaging indicates the generation of, on average, six times more FR-microbubbles with a wider

range of diameters (from 4 to 68 μm) that attach to the cells compared with NT-microbubbles (from 1 to 7 μm), with non-specific binding after activation. It is also found that the mean size of attached vaporised FR-nanodroplets is about three times larger than that of attached vaporised NT-nanodroplets. This investigation and quantification provide a better understanding of the acoustic response and size distribution of these vaporised targeted nanodroplets for future ultrasound molecular imaging and therapeutic studies.

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SUPPLEMENTARY MATERIALS

Supplementary material associated with this article can be found in the online version at doi:10.1016/j.ultrasmedbio.2019.01.009.

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